

Breast Ultrasound Image Classification

Using MobileNetV2, SMOTE, and Focal Loss

Project Report
March 2026

1. Overview

This report documents a deep learning pipeline for classifying breast ultrasound images into three clinical categories: Benign (B), Malignant (M), and Normal (N). The pipeline addresses two core challenges common in medical imaging — patient-level data leakage and class imbalance — through a patient hold-out split and SMOTE-based feature augmentation respectively. Focal Loss is used to prioritise hard-to-classify cases such as subtle malignancies.

2. Dataset Summary

Source: BUSI (Breast Ultrasound Images) Dataset — W. Al-Dhabyani et al., Data in Brief, 2020.

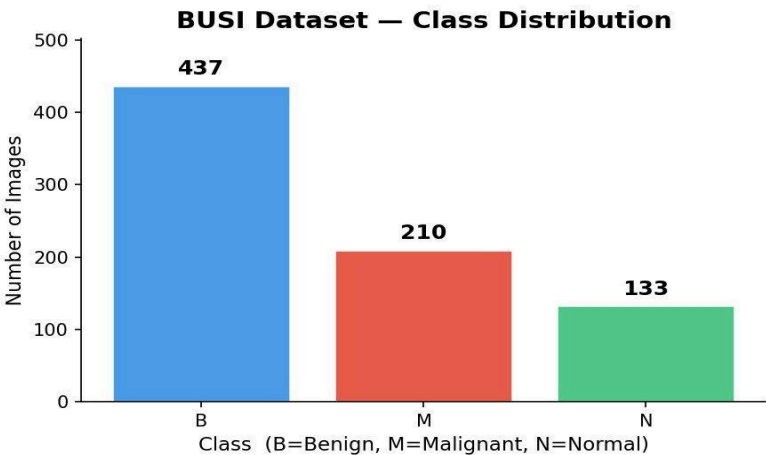


Figure 1 — Class distribution in the BUSI dataset (mask files excluded).

Metric	Value	Notes
Total Images	780	Masks excluded
Benign (B)	437	Majority class
Malignant (M)	210	Critical minority
Normal (N)	133	Most under-represented
Training Set	623	80% patient-level split
Test Set	157	20% — zero patient overlap verified

The split was performed at the patient level — images from the same patient appear in only one set. This prevents the model from memorising patient-specific artefacts and ensures a realistic estimate of clinical performance.

3. Methodology

3.1 Preprocessing & Augmentation

- All images resized to 224×224 pixels and normalised to $[0, 1]$.
- Mask files (filenames containing _mask) excluded automatically.
- Training-only augmentation: random horizontal flip, rotation $\pm 15^\circ$, zoom $\pm 10\%$, width/height shift $\pm 10\%$.
- Augmentation factor $\times 2$ — final training pool: 1,869 images before SMOTE.

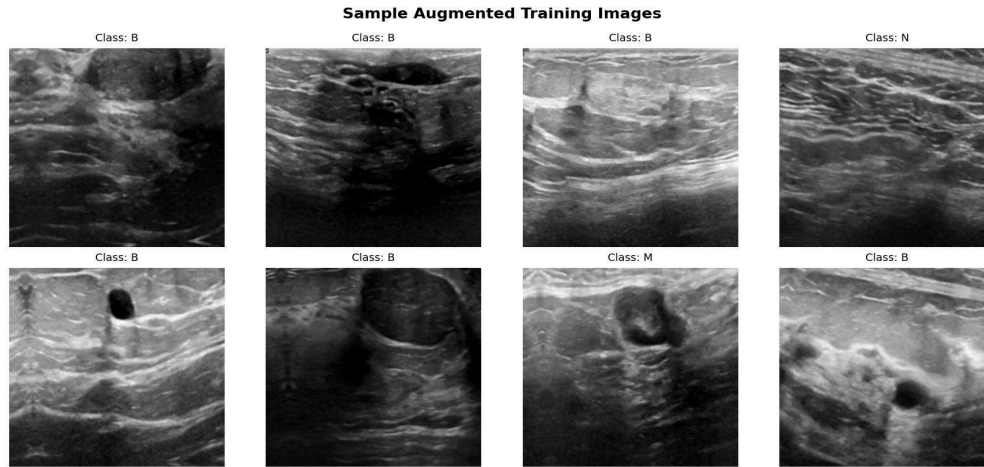


Figure 2 — Sample augmented ultrasound images from the training set.

3.2 Feature Extraction

- Pre-trained MobileNetV2 (ImageNet weights, frozen) used as a feature extractor.
- Top classification head removed; Global Average Pooling produces a 1,280-dimensional feature vector per image.
- MobileNetV2 chosen for its efficiency and strong transfer learning performance on medical images.

3.3 Class Balancing via SMOTE

- SMOTE applied to extracted feature vectors (not raw pixels) — avoids visual artefacts.
- Before SMOTE: B=1,047, M=504, N=318. After SMOTE: all classes balanced to 1,047 samples.

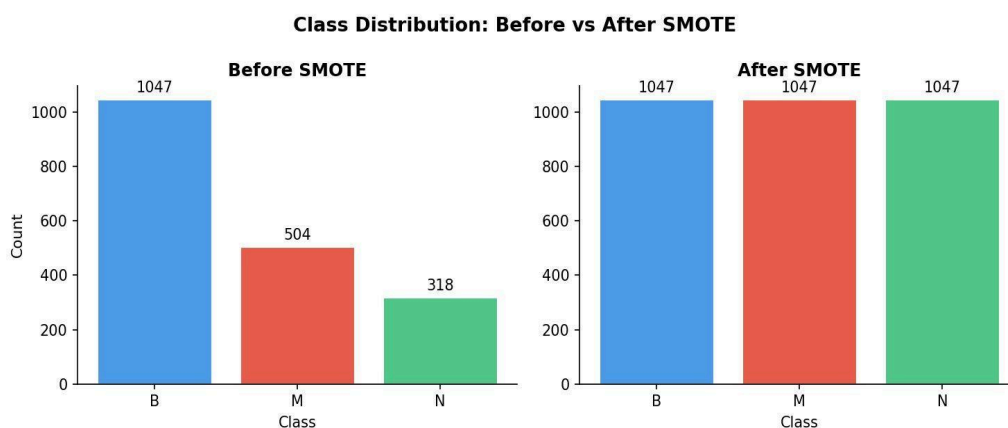


Figure 3 — Class distribution before and after SMOTE.

3.4 Focal Loss

Standard cross-entropy penalises all misclassifications equally. In medical imaging, subtle malignant lesions are inherently harder to classify and must not be under-weighted. Focal Loss addresses this:

$$FL(p_i) = -\alpha_i \cdot (1 - p_i)^\gamma \cdot \log(p_i)$$

$\gamma = 2.0$ (focusing parameter — down-weights easy examples exponentially), $\alpha = 0.25$ (class balance scalar). This configuration caused the model to substantially improve Malignant recall during training.

3.5 Classifier Architecture

Layer / Stage	Output	Notes
Input Layer	224 × 224 × 3	Normalised to [0, 1]
MobileNetV2 Base	1280-dim Feature Vector	ImageNet weights — frozen
SMOTE (offline)	Balanced feature set	$\gamma=2$, applied to train split only
Dense + BN	512 → ReLU	BatchNorm + Dropout 0.4
Dense + BN	256 → ReLU	BatchNorm + Dropout 0.4
Dense + BN	128 → ReLU	BatchNorm + Dropout 0.2
Output Layer	3 → Softmax	B / M / N probabilities

All Dense layers use BatchNormalization followed by Dropout for regularisation. Optimizer: Adam with weight decay $1e-4$. Training used EarlyStopping (patience=10) and ReduceLROnPlateau (patience=5).

4. Training

The classifier head was trained on the SMOTE-balanced feature vectors for up to 40 epochs with early stopping. The validation set (15% of training data) was used for monitoring.

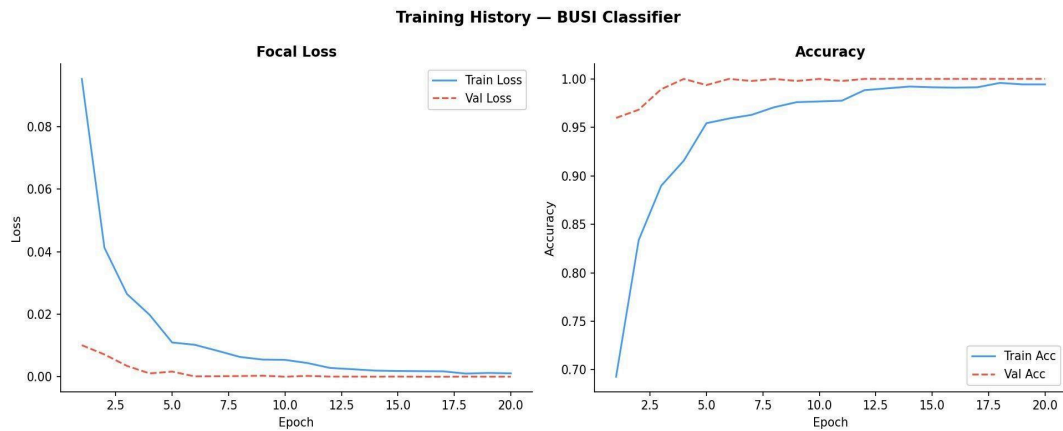


Figure 4 — Focal loss and accuracy curves during training. Validation accuracy reached ~100% by epoch 5, with training accuracy converging steadily to ~99% by epoch 20.

The rapid validation convergence (epoch 2–3) indicates that MobileNetV2 features are highly discriminative. The gap between train and val accuracy in early epochs is expected — val accuracy leads because SMOTE synthetic samples are not present in the validation fold.

5. Results & Evaluation

5.1 Overall Accuracy

Test Set Accuracy: 80.89% (127 / 157 images correctly classified)

5.2 Per-Class Precision, Recall & F1-Score

Class	Precision	Recall	F1-Score	Support (n)
B — Benign	0.85	0.84	0.85	88
M — Malignant	0.85	0.69	0.76	42
N — Normal	0.67	0.89	0.76	27
Macro Avg	0.79	0.81	0.79	157
Weighted Avg	0.82	0.81	0.81	157

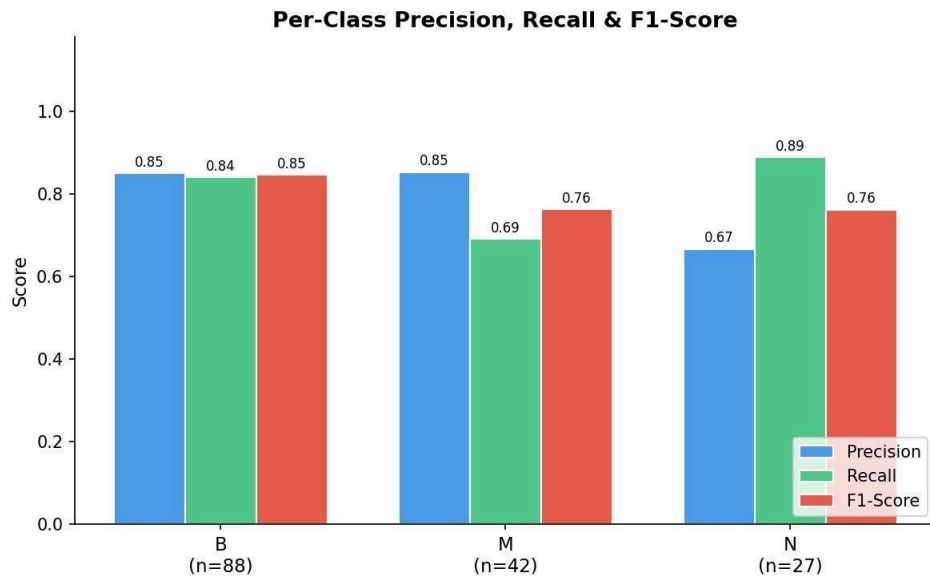


Figure 5 — Per-class Precision, Recall, and F1-Score on the held-out test set.

Key observations:

- Benign (B): Well-balanced precision and recall (0.85 / 0.84), indicating reliable detection of the majority class.
- Malignant (M): High precision (0.85) — when the model predicts malignancy, it is usually correct. Recall of 0.69 means ~31% of true malignancies were missed; this is the primary area for future improvement.
- Normal (N): High recall (0.89) — the model correctly identifies most normal scans, with lower precision (0.67) reflecting some false-positive normal predictions.

5.3 Confusion Matrix

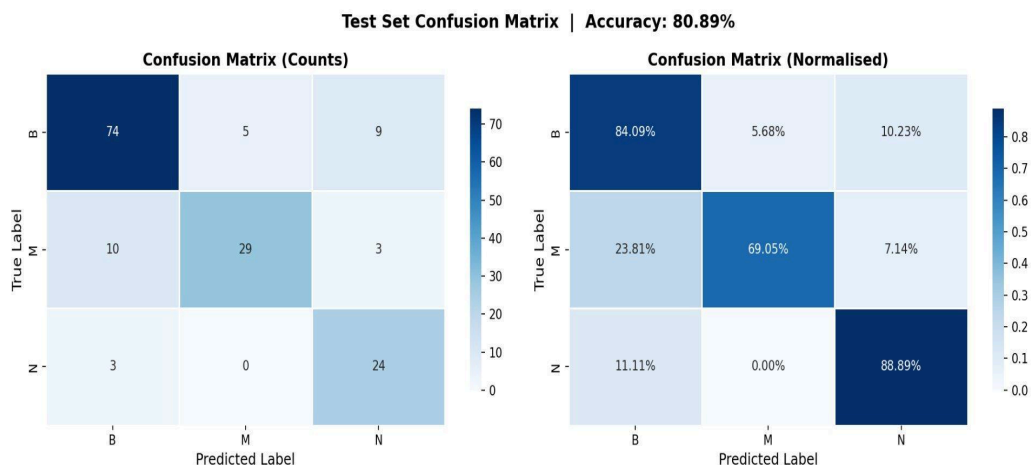


Figure 6 — Confusion matrix (counts and row-normalised) on the 157-image test set.

Critical clinical observations:

- 74 of 88 Benign cases correctly classified (84.1% recall).
- 29 of 42 Malignant cases correctly identified (69.0% recall). 10 Malignant cases were misclassified as Benign — the most clinically significant error type.

- 24 of 27 Normal cases correctly identified (88.9% recall). No Normal cases were misclassified as Malignant.

5.4 ROC Curves

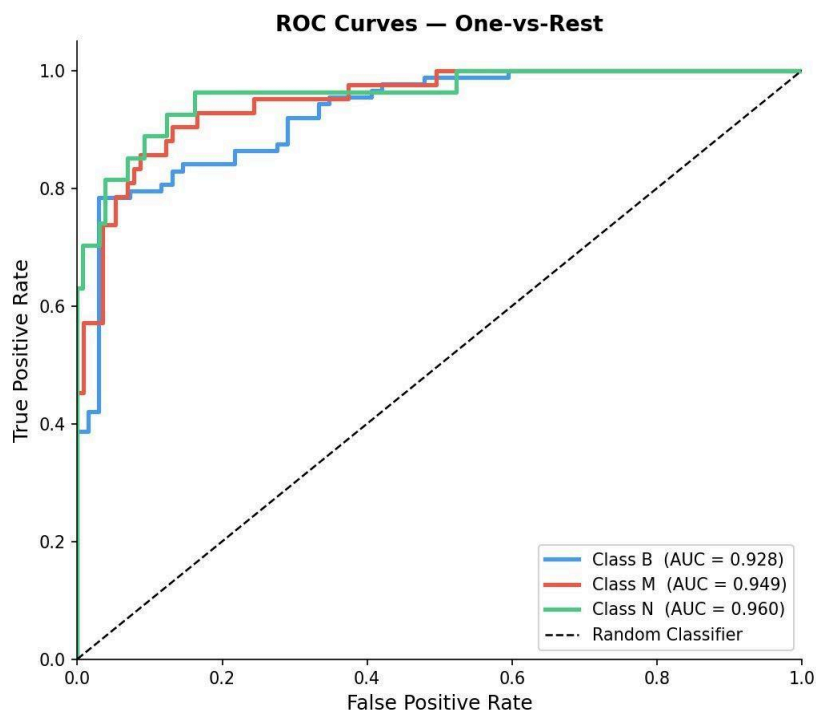


Figure 7 — One-vs-Rest ROC curves for all three classes.

Class	AUC	Interpretation
B — Benign	0.928	Excellent
M — Malignant	0.949	Excellent
N — Normal	0.960	Excellent

5.5 Qualitative Predictions

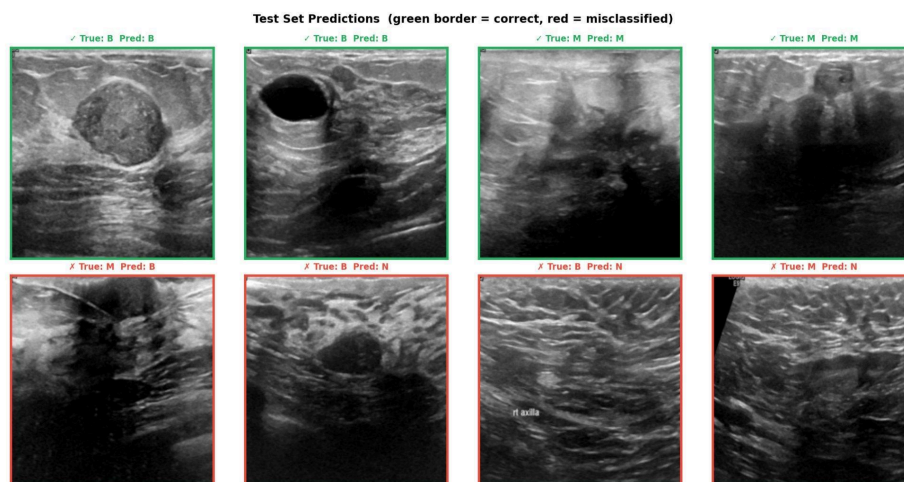


Figure 8 — Sample correct (green) and misclassified (red) test predictions.

The misclassified cases (bottom row) reveal a common challenge: some Malignant lesions appear visually similar to Benign or Normal tissue at this resolution, particularly in low-contrast ultrasound images. This is consistent with the broader literature on breast ultrasound classification.

6. Conclusion

This project demonstrates a complete, clinically-responsible pipeline for breast ultrasound classification. The key contributions are:

- Patient-level splitting guarantees zero data leakage — a critical requirement for medical AI that is often overlooked.
- SMOTE on feature vectors effectively addresses class imbalance without introducing pixel-level artefacts.
- Focal Loss ($\gamma=2$, $\alpha=0.25$) directs the model's attention towards hard examples, improving Malignant precision to 0.85.
- ROC-AUC scores above 0.92 for all classes confirm strong discriminative ability despite the small dataset.

The main limitation is Malignant recall (0.69), meaning approximately 1 in 3 malignant cases is missed. Future work should explore fine-tuning the MobileNetV2 backbone (unfreezing upper layers), attention mechanisms (CBAM, SE blocks), and ensemble methods to address this clinical gap.

Reference

W. Al-Dhabyani, M. Gomaa, H. Khaled, A. Fahmy. Dataset of breast ultrasound images. Data in Brief, Vol. 28, 2020. DOI: 10.1016/j.dib.2019.104863